



COE 158 Introduction to IT

History of Computers



Lecture Outline

- Evolution of Computer Hardware
 - Before the Generation
 - First Generation
 - Second and Third Generation
 - Fourth Generation
- Evolution of Computer Software
- Evolution of Computer User
- Impact on Society



Computer vs Other Human Inventions

- When we examine the role of the computer in our lives, we find that it plays a part in nearly everything we do
 - Communication
 - Commerce
 - Transportation
 - Education
 - Manufacturing
 - Research, etc
- What other human inventions have had the same impact?
 - Language
 - Taming (controlling) fire
 - Electricity
 - Refrigeration
 - The automobile
 - The airplane



Computers from 1950s to Today

1950s	Today
Cost: hundreds of thousands of \$ to millions of \$	\$300 - \$1200 for normal (desktop/laptop) computers
Thousands of instructions per second	Billions of instructions per second (supercomputers > 1 trillion per second)
Little main memory (a few thousand bytes?)	4-8 Billion bytes or more
Very large machines	Can be carried in your pocket or your brief case
Few users (mostly the engineers and programmers)	Billions of users

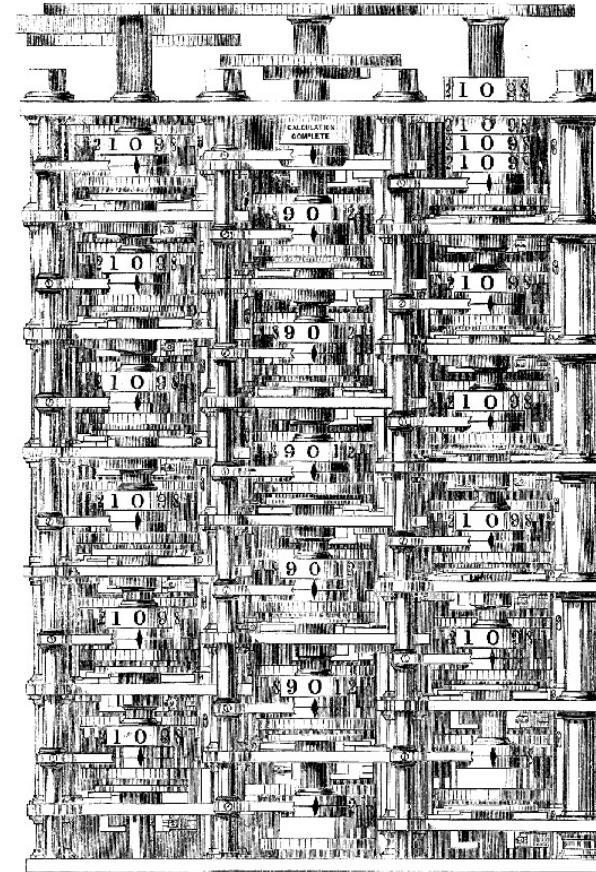
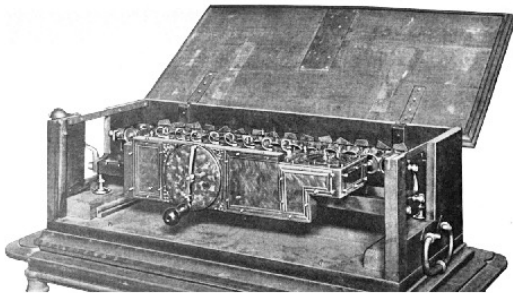
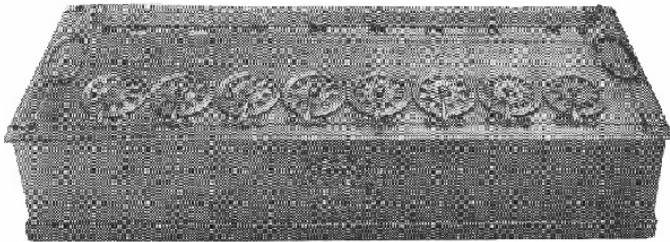
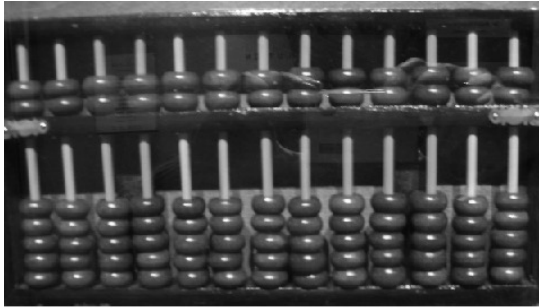


Before the Generations

- We describe computer technology by generation
 - We are currently in the 5th generation
- Prior to the 1st generation, we still had computational devices
 - The abacus (2000-4000 BC?), unknown inventors
 - used for simple counting
 - Mechanical calculators
 - Pascal and Leibniz (early to mid 1600s)
 - Programmable device
 - Jacquard's Loom (circa 1800)
 - Babbage's programmable devices
 - Difference engine (1822)
 - Analytical engine (1832)
 - neither device was completed by Babbage at the time as he ran out of funds



Abacus, Calculators, Loom, Analytical Engine



POSITION OF BABAGE'S DIFFERENCE ENGINE.



More on Babbage

- Device to compute mathematical tables
 - Use difference equations
 - Steam powered to rotate stalks of gears
 - Rotation of gears performs + and –
- Analytical engine programmable for different types of equations – IPO cycle, first computer!
 - Input equation
 - Process
 - Output results (via printing press)
 - Storage results (orientation of gears)
- Initial programs written by mathematician Lady Ada Lovelace (world's first programmer)



The 1930s-1940s

- Leading up to the first generation
 - Electricity used instead of steam power
 - Some electronic circuits replace mechanical computing
 - Not general purpose – computers could compute one type of operation, e.g., code breaking, rocket trajectories
- Circuits used switches and possibly vacuum tubes
- Some of these early devices were still analog in nature (not using 1s and 0s)
 - Slow, expensive, unreliable
- Demand for computers came from World War II



Notable Early Computers

Name	Year	Nationality	Comments
Zuse Z3	1941	German	Electromechanical, programmable
Atanasoff-Berry	1942	US	Electronic, non-programmable
Colossus Mark 1	1944	UK	Electronic, programmable
Harvard (Mark 1)	1944	US	Decimal, electronic, programmable
Colossus Mark 2	1944	UK	Electronic, programmable
Zuse Z4	1945	German	Electromechanical, programmable
ENIAC	1946	US	Electronic, programmable

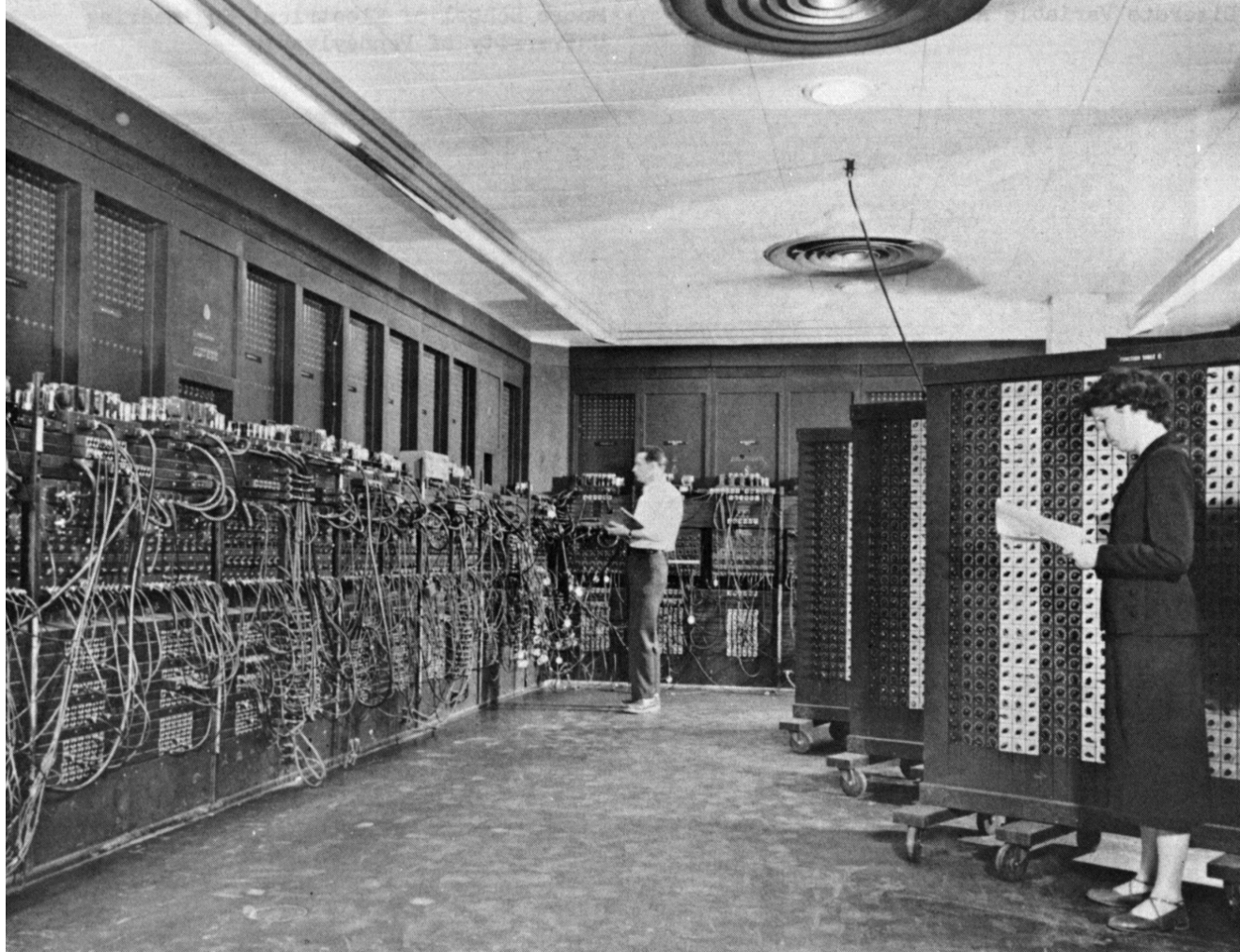


First Generation

- ENIAC (Electronic Numerical Integrator and Computer), Uni. of Pennsylvania, Feb 1946, \$500K
 - First general purpose programmable electronic computer
 - Weighed 30 tons, about 1800 square feet, 17,468 vacuum tubes, 7200 crystal diodes, 1500 relays, 70,000 resistors, 10,000 capacitors, millions of hand-soldered joints
 - Input from punch cards, output using an offline accounting machine for a printer, magnetic tape used to move output
- Most first generation computers were one-of-a kind laboratory machines
 - Experimental, large, costly, slow, unreliable because of vacuum tubes, programmed using machine language
 - Used only by the engineers and technicians of the lab
 - By the mid 1950s, computer companies were selling mainframe computers to those who could afford them



The ENIAC



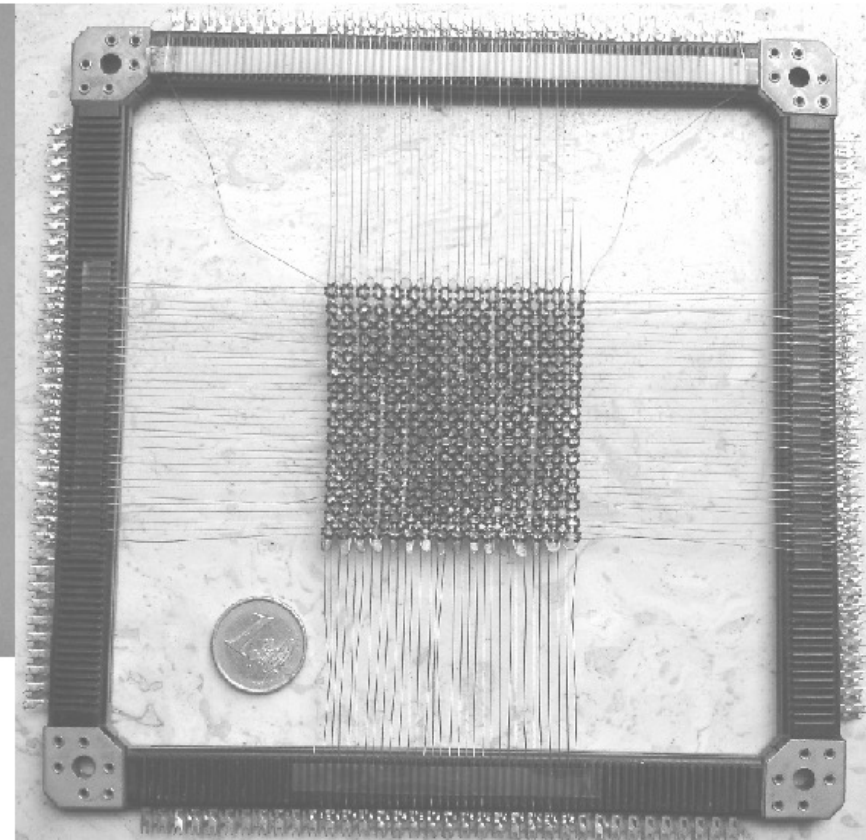
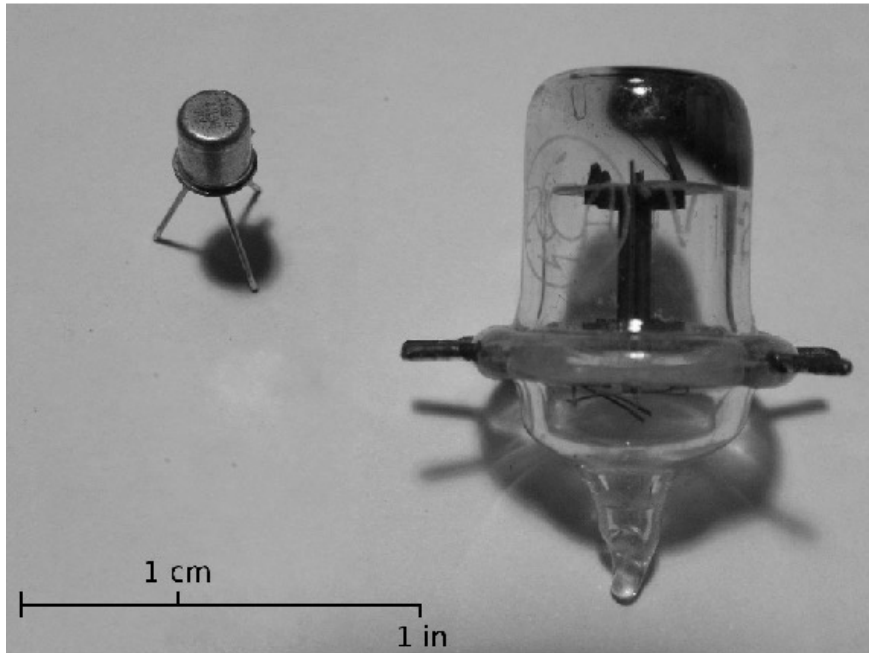


Second Generation

- Started around 1959
 - Transistors replaced vacuum tubes
 - Magnetic core memories introduced for memory storage
 - expensive but permitted up to 1MByte of storage
 - Disk drum (data drum) introduced for storage
- Smaller computers meant faster computers
 - Less distance for current to travel
- Cheaper computers
 - Components were less costly, computer companies were formed, computers were no longer “1 of a kind”
- Led to increased number of computers and therefore increased usage
- Need for software support to run programs
 - the resident monitor



Vacuum Tube, Transistor, Magnetic Core Memory





Third Generation

- Started around 1963 (or as late as 1965)
- Circuits placed on silicon chips (integrated circuits) were used to replace transistors (logic) and mag core memory
- Made computers smaller, faster, cheaper, more easily produced
- Led to more computers being manufactured and purchased
 - Computer users were no longer (necessarily) programmers/engineers or even highly trained
 - Resident monitor upgraded to full operating system
 - Networks and dumb terminals permitted access remotely (from outside of the computer room)
- Minicomputers introduced (scaled down mainframes, cheaper, not as powerful) allowed smaller organizations to own computers
- Computer families introduced

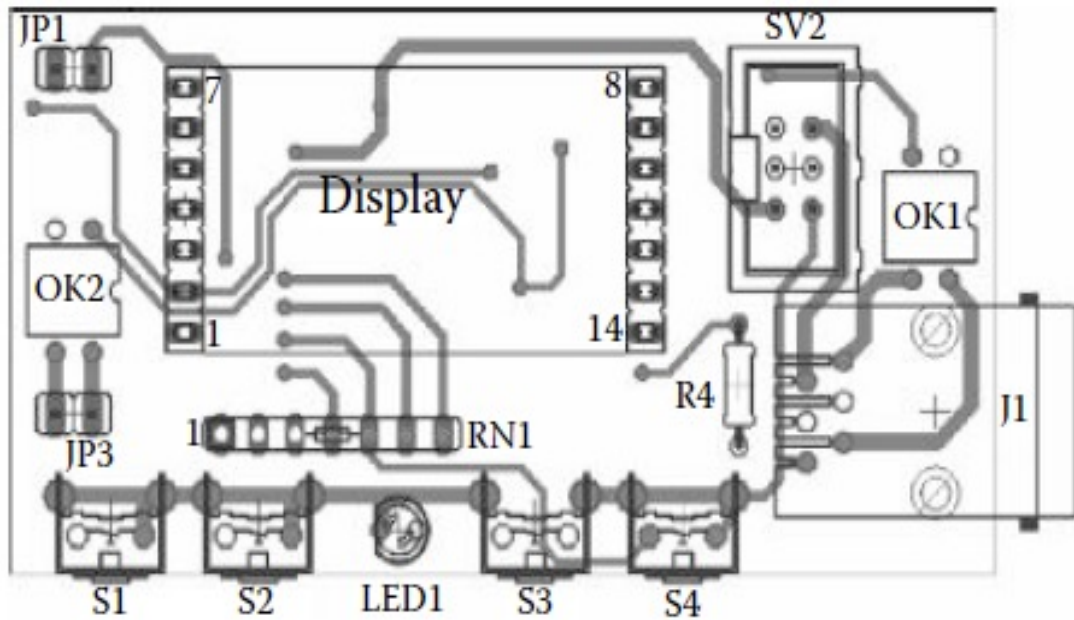


Fourth Generation

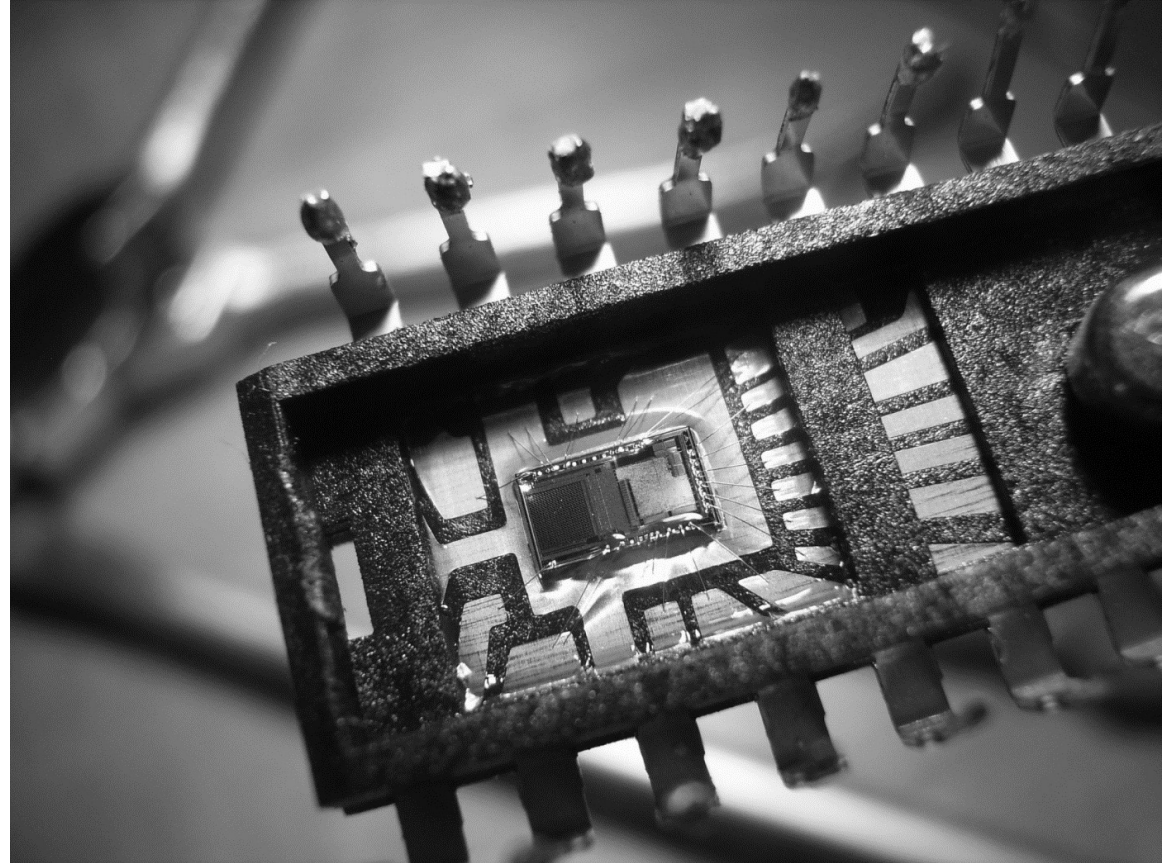
- Started as early as 1971 (or maybe 1974)
- Miniaturization led to CPU on one chip
 - The transistor count has increased exponentially over time, roughly doubling every 18-24 months
 - SSI – 10-20 (early 60s) MSI – 100s (mid to late 60s)
 - LSI – 1000s (early 70s) VLSI – 100,000s (late 70s)
 - ULSI – millions+ (80s and beyond)
 - 2012: up to 2.5 billion transistors on a CPU
 - Led to the creation of the microprocessor
- Microprocessor led to the first computers called microcomputers
 - The first personal computers introduced in the 1970s, largely hobbyist toys, little main memory, few software titles, I/O largely limited to keyboard, printer or tv set, cassette drive



Chip with Miniaturized Circuitry



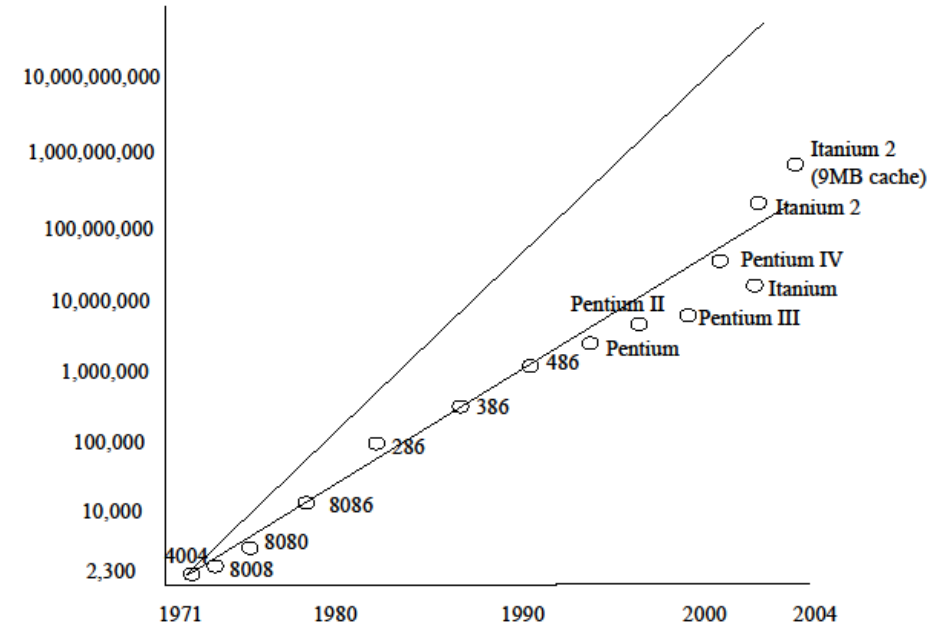
An integrated circuit





Moore's Law

- Gordon Moore, one of the founders of Intel
 - Noticed the increased transistor count as early as 1965
 - Used it to predict transistor count for the future
 - Doubling of transistors every 18-24 months
- If we explore microprocessors, we see that Moore's Law has come true since 1970 until present day! (see the next slide)
 - This is not a physical law, it is an observation and a prediction



Computer engineers predict that we will no longer be able to fulfill

Moore's Law

hasn't happened yet

but might happen any year now



Other 4th Generation Inventions

- Laptop (and smaller) computers using LCD screens, long lasting batteries
- Laser printers, touch screens
- Broadband wireless network access
- Gigabyte memory storage, terabyte disk storage
- CPU innovations
 - Pipelined CPUs
 - Parallel processing
 - Multicore processors
 - On-chip and off-chip cache memories
 - Register windows
 - Branch speculation



*Fifth Generation (Present and Beyond)

- **Fifth generations computers are only in the minds of advance research scientists and being tested out in the laboratories.**
- These computers will be under Artificial Intelligence(AI), They will be able to take commands in an audio visual way and carry out instructions.
- Many of the operations which requires low human intelligence will be performed by these computers.
- Parallel Processing is coming and showing the possibility that the power of many CPU's can be used side by side, and computers will be more powerful than those under central processing.
- Advances in Super Conductor technology will greatly improve the speed of information traffic. Future looks bright for the computers.



Milestones Comparisons

Year	Automotive Milestone	Computing Milestone	Year
6500 B.C.	The wheel	Abacus	4000? B.C.
1769	Steam-powered vehicles	Mechanical calculator	1642
1885	Automobile invented	Programmable device	1801
1896	First automotive death (pedestrian hit by a car going 4 mph)	First mechanical computer designed (analytical engine)	1832
1904	First automatic transmission	First digital, electronic, general purpose computer (ENIAC)	1946
1908	Assembly line permits mass production	Second-generation computers (cheaper, faster, more reliable)	1959
1911	First electric ignition	Third-generation computers (ICs, cheaper, faster, computer families)	1963
1925	About 250 highways available in the United States	ARPANET (initial incarnation of the Internet)	1969



Milestones Comparisons...

Year	Automotive Milestone	Computing Milestone	Year
1940	One quarter of all Americans own a car	Fourth-generation computers (microprocessors, PCs)	1974
1951	Cars reach 100 mph at reasonable costs	First hard disk for PC	1980
1956	Interstate highway system authorized (took 35 years to complete)	IBM PC released	1981
1966	First antilock brakes	Macintosh (first GUI)	1984
1973	National speed limits set in the United States, energy crisis begins	Stallman introduces GNUs	1985
1977	Handicapped parking introduced	Internet access in households	1990s
1997	First hybrid engine developed	Cheap laptops, smart phones	2000s



Evolution of Computer Software

- Programming language evolution
 - Machine language (1st generation)
 - Assembly language (1st and 2nd generations)
 - High level languages (2nd generations and onward)
 - Structured Programming (3rd and 4th generations)
 - Object-oriented languages (4th generation)
 - Visual programming (4th generation)

C code

```
for(i=1;i<=x;i++)  
    sum=sum+i;
```

Assembly Language code

```
01231407    mov     dword ptr [i], 1  
0123140E    jmp     main+69h (1231419h)  
01231410    mov     eax, dword ptr [i]  
01231413    add     eax, 1  
01231416    mov     dword ptr [i], eax  
01231419    mov     eax, dword ptr [i]  
0123141C    cmp     eax, dword ptr [x]  
0123141F    jg      main+7Ch (123142Ch)  
01231421    mov     eax, dword ptr [sum]  
01231424    add     eax, dword ptr [i]  
01231427    mov     dword ptr [sum], eax  
0123142A    jmp     main+60h (1231410h)
```

Machine Language code

```
C7 45 EC 01 00 00 00  
EB 09  
8B 45 EC  
83 C0 01  
89 45 EC  
8B 45 EC  
3B 45 F8  
7F 0B  
8B 45 E0  
03 45 EC  
89 45 E0  
EB E4
```



Evolution of the Computer User

- 1950s – users were engineers and programmers
 - They wrote programs and the code needed for I/O
 - Tested them out, debugged the code and fixed the hardware
- 1960s and 1970s – users were highly trained employees but not necessarily programmers
 - Used operating system commands to run their “jobs” including for instance JCL commands in IBM mainframes and VMS commands on DEC’s Vax computers
- 1980s – PCs became available
 - Simple DOS commands, later GUIs
- 1990s up to today – point and click, look and feel



Impact on our Society

- Our parents (our grandparents) generation used to think of computers as
 - Scary, unknowable, threatening, big brother
- Today, nearly everyone on the planet has Internet access
- When do you not use a computer in some aspect of your life?
 - Driving? GPS, digital radio, antilock brakes
 - News? Gathered and disseminated by computer, possibly read on-line
 - Movies? writing, editing, special effects, marketing, etc all involve computers
 - Communication? Cell phones, satellites, etc.
 - Shopping? credit card communication, store inventory, computerized cash registers
 - Going outside? Weather prediction modeling on computers



From Resident Monitor To OS

- Early computers had no OS
 - Programmers had to include code to handle all input/output and program startup
 - Resident monitor alleviated need for some of these duties
- Resident monitor developed starting in the 1950s to handle punch card instructions for I/O
- OS developed out of resident monitors by adding more and more utilities and support for the user
 - This evolution started in the late 1950s and continues today
 - Most early operating systems were developed to support only one hardware platform



Noteworthy Operating Systems

- GM-NAA I/O – General Motors for IBM 701, collection of resident monitors to support the user, 1955
- Atlas Supervisor – Manchester University, first true OS, permitted concurrent user access and virtual memory
- BESYS – Bell Labs, 1957 for IBM 704, handle input from punch cards and tape, output to printer or tape, supported FORTRAN programs
- IBSYS – for IBM 7090 and 7094, 1960, OS commands were embedded in programs using \$ to indicate OS command rather than program instruction (JCL would copy this format later)
- CTSS – MIT, 1961, first true time-sharing system
- EXEC 8 – Remington Rand's UNIVAC, 1964, supported batch, time sharing and real-time processing
- TOPS-10 – DEC PDP-10, 1964, time sharing with shared memory
- MULTICS – 1964, hardware independent through modules, dynamic linking of program code, access control lists, storing process memory like files, hierarchical file system
- OS 360 – IBM 360 (and later IBM 370), 1966, batch processing with threading within a single task, JCL commands for I/O, virtual memory, hierarchical file system, virtual storage access method (used for database access), parent/child spawning, network communications
- Unics – AT&T, 1969, an early precursor to Unix based on MULTICS



History of Unix

- Up until this time, most OSs were developed for specific hardware
- One goal behind Unix was to be portable
 - Hardware independent, or an OS that could run on multiple platforms
- Features to include
 - File system administration
 - Strong network components, TCP/IP
 - Security
 - Custom software installation
 - Define your own kernel programs
 - Shells, scripting
 - Command line interpreter, redirection and pipes

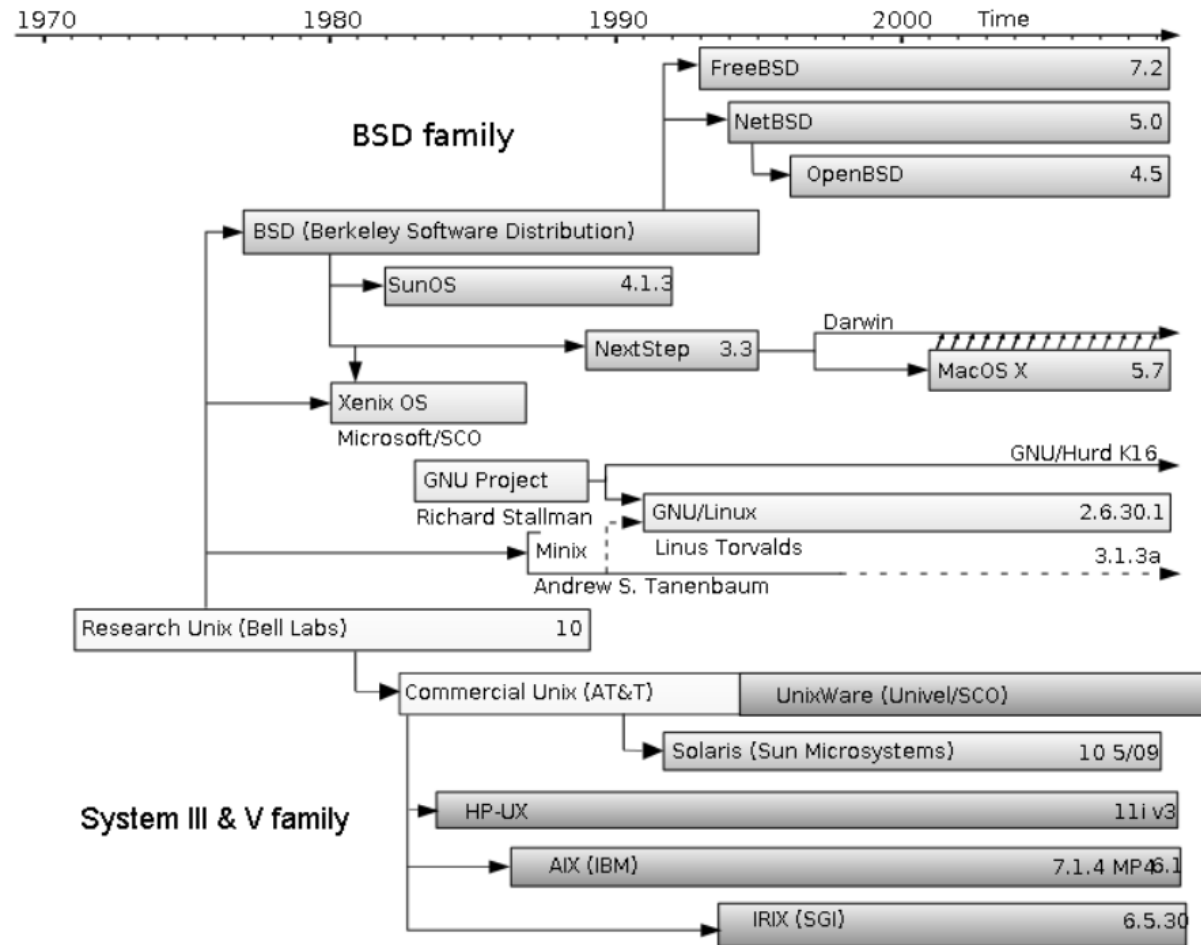


Unix/Linux Timeline

- Up until the mid 1970s, Unix was only available commercially (for sale)
 - Thompson, in a sabbatical at UC Berkeley developed the Berkeley Standard Distribution (BSD) which would compete with Bell Labs' Unix
 - BSD spawned SunOS, FreeBSD, NetBSD and eventually MacOS X
 - Bell Labs Unix spawned AT&T Unix, Solaris (Sun), HP-UX, AIX (IBM) and others
 - GNU's was a project developed by Richard Stallman at MIT
 - 1991 – Linus Torvalds, unhappy with his OS class' OS begins developing Linux, asking OS programming community for help
 - 1993 – first major distribution of Linux released



Unix/Linux Timeline





Linux Distributions

- Initial development of Linux kernel

- MCC Interim Linux/TAMU Linux (Texas A&M)

- DLD

- SLS/Slackware

- Debian Linux

- Free BSD

- S.u.S.E.

- Red Hat

- Knoppix

- Mandrake

- Yellow Dog

- Enoch

- Arch Linux

- Oracle Linux

- Fedora

- CentOS

- Morphix

- PC Linux

- Ubuntu Lite

- Ubuntu

- Linux Mint

- Kanotix

- Kubuntu

- Mandriva

- Mageia

- OpenSUSE

1991 1993 1995 1997 1999 2001 2003 2005 2007 2009 2011



Popular Linux Distros

- Two most popular distributions are Red Hat and Debian
 - Both have splintered to create many sub-distributions
- Red Hat – both open source and commercial versions available, SELinux, Red Hat Package Manager (rpm)
 - Fedora
 - CentOS
- Debian – completely open source (not commercially available), has over 20,000 software packages available for it, uses APT package manager, can boot from live CD
 - Ubuntu
- SLS/Slackware – German version of Linux, bought by Novel in 2003
 - SuSE



Free Software Movement

- Started by Richard Stallman as part of the GNU's project
 - GNUs – GNUs not Unix – a joke by Stallman to define his project using a recursive definition
 - He attempted to enlist help in the development of GNUs – a completely open source Unix-like OS
 - The goal was to build an open source operating system
- Free as in freedom, not beer
 - Software are ideas and ideas should not be owned, so all software should be made available not just for free but in source code so that code could be shared and developed
- Others embraced the open source initiative, they did not necessarily think that all programming endeavors should be free



Open Source and GPL

- Stallman felt all software should be free, particularly anything developed out of other free software
- Torvalds adopted an attitude that Linux and the other software created for Linux
 - can be available in open source
 - can be sold if the programmer chooses
- This led to a split
 - Free Software Foundation – Stallman's group
 - Open Source Movement – those who work in the open source community but permit commercial products to be built from partially open source software
- Stallman defined the GNU's Public License
 - Anything created with open source tools must be published using the GPL
 - Stallman called this a “copyleft” instead of a copyright
 - Both groups use the GPL (most of Linux is published under the GPL)
- The GPL is now used in other forms of publishing
 - see for instance wikimedia



Pre-Windows PC History

- MS-DOS – released in 1981 for the IBM PC
 - IBM published their architecture and used off-the-shelf components
 - This led to “clones” (or compatibles)
 - Since software available for IBM PC would run on clones, these computers captured a larger marketshare
 - MS-DOS ran on all of these
 - Thus, MS-DOS was very popular
- MS-DOS
 - Single tasking
 - Text-based
 - Most commands revolved around file operations (DOS = disk operating system)



Xerox PARC → Macintosh

- Xerox Palo Alto Research Center
 - Artificial intelligence research, programmers implemented windows based operating system in the 1970s using object-oriented programming
 - They offered tours of their facility where they showed off their operating system (Jobs and Wozniak of Apple saw this when they took a tour of Xerox Parc in the late 70s)
- With the success of the Apple I and II personal computers, Jobs and Wozniak started development of other projects
 - Jobs project was Apple Lisa but he was kicked off the project when it ran into delays and went over budget!
- Lisa released first (1983), too expensive to be successful (\$10K/computer), Mac released in 1984, was very successful



Microsoft Windows

- During the development of the Macintosh and Lisa, Apple hired Microsoft to develop software for the Macintosh
 - Supposedly, this led to Bill Gates designing and implementing Windows
- Windows 1.0 released in 1985 ran on top of MS-DOS
- Windows 2.0 released in 1987, 3.03 in 1990, 3.1 became very popular
- Windows NT – networking based version, 32-bits, supported client-server networks and competitive multitasking
- Windows 95 became the first PC (stand-alone) OS from Windows to support multitasking and networking, also introduced plug-and-play
 - Windows 98, Windows 2000, Windows ME and Windows XP, Windows Server (2003), Windows Vista in 2007 was a failure, Windows 7 (2009), Windows 8 (late 2012), Windows 10 (2015)



Mac Developments

- Mac
 - Multitasking for background processes in 1987, Virtual memory support in 1991, Desktop shortcuts in 1991, TCP/IP networking in 1999, Unix-like kernel in 2001
- In an ironic twist, Microsoft in 2005 invested heavily in Apple to help support the struggling company
 - Their condition: change from the PowerPC processor to the Intel processor so that Microsoft software would run on Macintosh
 - Mac hardware had always been superior to PC hardware but not any longer
 - The more expensive hardware was partially why Macintoshes have been more expensive